

## Lab: Mess With DNS

Messwithdns.net is a domain that's just for playing around that lets anyone set DNS records on it.

The project was built by Julia Evans and Marie Claire LeBlanc Flanagan who says, "the best way to learn

is to experiment! Do weird DNS experiments and see how it works." It is a fun way to gain experience

creating and troubleshooting DNS entries. This is my experience experimenting with DNS:

Tutorial experiments: These 3 experiments explain some DNS basics:

### 1. Create an A record

I set up my first A record, and when I clicked the link, a page with an orange background popped up.

### 2. Create a CNAME record

I made a CNAME record or an alias record so that when I type fruit plus my subdomain, I still get sent to the same page. Although the domain name was changed slightly (subdomain), I was still sent to the same page.

### 3. See how CNAME records work

CNAMEs are really cool, as you can change the way you type a domain name and still get taken to the same page. Take Google, for example. If you type [google.com](https://google.com) or google, you still get taken to the same search engine. It makes it easier for the user to remember domain names, especially if the domain names get really complicated.

Weird experiments: In these 5 experiments, we break things for fun and learning!

### 1. Visit a domain before creating its DNS record

I created the DNS after visiting the webpage and got a server not found sign. This happened because of Negative caching. What essentially happened was that by my visiting the site before the DNS was up, the server remembered that the webpage was not in its database, so when DNS asked again for the webpage, it remembered that the page doesn't exist even though it did!

### 2. Set a very long TTL

When setting the TTL(time to live) for a long duration the webpage is cached in the DNS servers so when you try to reload the new URL it stayed to the previous page. In this case the previous page was a page with an orange in the background. The new page was supposed to be a page with some grapes in the background. However since the the TTL was so long the DNS kept the domain name to the page with the orange and didn't update to the new one with a grape.

### 3. Point to example.com

I made 2 domains share the same IP address. I put in the real domain name with the an IP address of 93.184.216.34 and I made a new domain with a similar domain but not exactly the same. When trying out the new domain which shares the same IP I do not get the same web page. It is because the server doesn't recognize the name so it sends a 404 message.

### 4. Convince 3 different DNS servers that your subdomain has 3 different IPs (3-ips)

I made 3 DNS think I had 3 IP's for the same subdomain. The reason it worked was because I made the time to live(TTL) 3600 seconds. This was enough time for the 3 servers to cache my IP address that I set up. So they each have different variations on my IP address because the

TTL did not expire fast enough to for the servers to update. Since they look at the cache memory first they see the IP address that I gave them and just recite it back to the host ie me.

#### 5. Set two conflicting A records

I set up 2 A records with the 2 different IP's. The subdomains are the same but the IP's are not. I set the time to live to 5. Since the refresh rate is so high the same domain gives me 2 separate web pages. So it goes back and forth on the web pages because they are both correct they are just pointing to 2 separate web pages.